OCT
15

Debug the System: ATL : Critical thinking: Analysing, Debugging, and Troubleshooting Systems
Chaitra R
Formative Formative 16/18 Students Thursday at 10:05 AM
IB MYP Design - Ms Ishita (MYP2) B

Task Details

You are given a **faulty circuit + code** that is **not working as expected**. Your role is to act as a **troubleshooter**, analyse them carefully, identify errors, and fix them logically.

This task is **not about getting the right answer quickly**, but about **thinking critically**, testing ideas, and justifying decisions.

What will you do:

1. **Observe the Problem**
    - Run the simulation as it is.
    - Record what *should* happen vs what *actually* happens.
2. **Break the System into Parts**
    - Identify inputs, outputs, and processes.
    - Separate **circuit issues** from **software (code) issues**.
3. **Identify Possible Errors**
    - List at least **3 possible reasons** why the system is not working.
    - Categorise errors (circuits, logic, values, conditions).
4. **Test & Debug**
    - Change **only one thing at a time**.
    - Test after every change.
5. **Fix the System**
    - Correct the identified errors.
    - Ensure the simulation now works as intended.
6. **Justify Your Thinking**
    - Explain:
        - What the error was
        - Why it caused the problem

- How your fix solved it

Task Specific Clarification

1–2

- Identify at least one error in the given circuit or code with guidance.

- Identify the input and output of the system.

- Make a simple correction and state what was changed.

3–4

- Describe what the circuit should do and what happens when the simulation runs.

- Identify at least two possible errors in the circuit or code.

- Make corrections and test whether the system works.

- Explain in simple words how the corrections helped.

5–6

- Identify the inputs, processes and outputs of the system.

- Identify at least three possible errors in the circuit or code and classify them as circuit or code errors.

- Change one thing at a time and test the simulation after each change.

- Correct the errors and explain why they caused the problem and how the fixes helped the system work as expected.

7–8

- Systematically investigate the circuit and code to identify and explain the causes of the errors.

- Identify at least three possible errors and distinguish between circuit connection errors, incorrect values and code logic errors.

- Follow a logical debugging process, changing one thing at a time and recording the results of each test.

- Correct the errors and test the system to confirm that it works as intended.

- Justify each correction by explaining the cause of the error, its effect on the system and why the solution works.

- Suggest one improvement that could make the circuit or code more reliable.